The Coherent Transition-Map Theorem

A Proposed Paper AE for the Dream-Paper Suite

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Abstract

Paper AD proves the Overlap Route-Agreement Lemma (AA3): licensed extraction procedures agree on overlaps up to a unique canonical **POT**-isomorphism. That result is the descent seed — it guarantees local coherence of the **POT**-fibration. But local coherence alone does not assemble local fiber data into a global directed diagram. For that, one needs canonical transition maps between fibers across all admissible enlargements, satisfying identity and composition axioms and respecting the overlap isomorphisms of Paper AD.

The present paper, Paper AE, proves the Coherent Transition-Map Theorem (claim AA4): for every admissible enlargement $\iota : R \hookrightarrow R'$, there exists a canonical **POT**-morphism $\tau_\iota : \mathcal{F}_R \to \mathcal{F}_{R'}$ satisfying identity, composition, and overlap-compatibility. Together with Paper AD, this produces a genuine directed diagram $\mathbf{D}_{\mathrm{pers}}$ in **POT** indexed by the base category **Reg**.

The proof constructs each transition map component by component using the reindexing functors of Paper AA and the kernel's admissible-test discipline. Canonicality is forced by the same argument that forced the ORA isomorphism: the fiber data is defined intrinsically at the regime level, and any structure-preserving map between fibers is uniquely determined by the source-fiber data.

This upgrade from Open to Unc for AA4 converts the **POT**-fibration from "locally coherent" to "globally directed." The persistence diagram $\mathbf{D}_{\mathrm{pers}}$ is now a mathematical object, and the question of whether it has a universal completion — the universal object $\mathcal{U}$ — becomes a legitimate construction problem.

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1 Purpose and imports

1.1 What this paper proves

Paper AE proves the Coherent Transition-Map Theorem, upgrading claim AA4 from OPEN to UNC, and produces the persistence diagram $\mathbf{D}_{\text{pers}}$ as a mathematical object in **POT**.

1.2 Imports

1. All of Papers A, B, D (kernel, LS-2, certificates).
2. Paper AA's definitions of **POT**, **Reg**, fibers, and reindexing functors.
3. Paper AD's Overlap Route-Agreement Lemma (AA3, now UNC).

1.3 What this paper does not prove

(i) Uniqueness of the transport invariant normalization (Paper AF, still OPEN);

(ii) the Amplified Minimal-Carrier Theorem (Paper AG, still OPEN);

(iii) the existence of the universal object $\mathcal{U}$ (AA7, still OPEN).

2 The transition map: construction

Definition 2.1 (Transition map data)**.** For an admissible enlargement $\iota : R \hookrightarrow R'$, the *canonical transition map* $\tau_\iota : \mathcal{F}_R \to \mathcal{F}_{R'}$ is defined component by component as follows.

Lemma 2.2 (Quotient transition)**.** *The admissible enlargement $\iota : R \hookrightarrow R'$ induces a canonical injective map $(\tau_\iota)_\Sigma : \Sigma_R \to \Sigma_{R'}$ between observational quotients, satisfying:*

1. *$[\omega]_R \mapsto [\iota(\omega)]_{R'}$ for $\omega \in \Omega_R$;*
2. *this map is injective (distinct R-classes map to distinct R'-classes); and*
3. *it commutes with the canonical isomorphisms of Paper AD for any compatible overlap: if $\iota = \iota_1$ and $\iota' = \iota_2$ share a common source Q, then $(\tau_{\iota_1})_\Sigma$ and $(\tau_{\iota_2})_\Sigma$ are compatible via $\theta_{R,R';Q}$.*

Proof. Well-definedness: if $\omega \sim_{\mathcal{A}} [\mathcal{A}_R]\omega'$ in R, then $\iota(\omega) \sim_{\mathcal{A}} [\mathcal{A}_{R'}]\iota(\omega')$ in R' by the admissibility condition (Paper AD, Definition 2.1(2)). So the map on equivalence classes is well-defined.

Injectivity: if $[\iota(\omega)]_{R'} = [\iota(\omega')]_{R'}$, then $\iota(\omega) \sim_{\mathcal{A}} [\mathcal{A}_{R'}]\iota(\omega')$. By the "only if" direction of Lemma AD-2.1 (quotient preservation), $\omega \sim_{\mathcal{A}} [\mathcal{A}_R]\omega'$ in R, so $[\omega]_R = [\omega']_R$.

Overlap compatibility: this is exactly Paper AD's Corollary AD-3.2 (consistent overlap data) applied to the quotient component. □

Lemma 2.3 (Continuation and ledger transition)**.** *The admissible enlargement ι induces a canonical map $(\tau_\iota)_\Lambda : \Lambda_R \to \Lambda_{R'}$ on continuation structures, satisfying:*

1. *every continuation history h on R maps to the corresponding history $\iota(h)$ on R' (same steps, same types, relabeled by ι);*

2. *the defect count satisfies* $\Lambda_{R'}(\iota(h)) = \Lambda_R(h) + \delta_\iota(h)$ *where* $\delta_\iota(h) \geq 0$ *counts any new defect events introduced by the enlargement (zero if* ι *is right-stable); and*

3. *the map commutes with the ORA isomorphisms on any compatible overlap.*

Proof. The continuation history h on R consists of a sequence of admissible extension steps on Ω_R. The map ι relabels each step to act on $\iota(\Omega_R) \subseteq \Omega_{R'}$. The resulting sequence is a valid continuation history on R' because ι is admissible (it preserves the test structure). The defect increment $\delta_\iota(h) \geq 0$ accounts for any step in the enlargement from R to R' that is defect-mediated; if the enlargement is right-stable, $\delta_\iota = 0$ and the defect count is preserved. Overlap compatibility follows from Paper AD's Lemma AD-3.2. □

Lemma 2.4 (Transfer and boundary-law transition)**.** *The admissible enlargement* ι *induces a canonical map* $(\tau_\iota)_T : \mathsf{T}_R \to \mathsf{T}_{R'}$ *on transfer structures. For each local region* $U \subseteq \Omega_R$*, the map sends the boundary-law data* $\Phi_R(U)$ *to the boundary-law data* $\Phi_{R'}(\iota(U))$*, compatibly with the LS-2 collar and outcome structure.*

Proof. The boundary-law map $\Phi_R : U \to O(U)$ for $U \subseteq R$ depends only on the stabilized collar type of U. Under ι, the region U maps to $\iota(U) \subseteq R'$, and the collar of $\iota(U)$ in R' includes the image of the collar of U in R, plus any new collar elements from $\Omega_{R'} \setminus \iota(\Omega_R)$. By the admissibility of ι, the new collar elements cannot create new distinctions among the existing collar classes (Paper D). So the collar quotient $\iota(U)$ in R' is canonically identified with U in R plus any genuinely new collar classes introduced by the extension. The map $(\tau_\iota)_T$ sends $\Phi_R(U)$ to the canonical extension $\Phi_{R'}(\iota(U))$. □

3 The Coherent Transition-Map Theorem

Theorem 3.1 (Coherent Transition-Map Theorem)**.** *For every admissible enlargement* $\iota : R \hookrightarrow R'$ *in* **Reg***, there exists a canonical* **POT***-morphism*

$$\tau_\iota : \mathcal{F}_R \longrightarrow \mathcal{F}_{R'}$$

satisfying:

(a) ***Identity****:* $\tau_{\mathrm{id}_R} = \mathrm{id}_{\mathcal{F}_R}$*;*

(b) ***Composition****:* $\tau_{\kappa\circ\iota} = \tau_\kappa \circ \tau_\iota$ *for any composable pair* $R \hookrightarrow R' \hookrightarrow R''$*;*

(c) ***Overlap compatibility****: for any compatible overlap* $\iota_1 : Q \hookrightarrow R$*,* $\iota_2 : Q \hookrightarrow R'$ *with common completion* R''*, the transition maps satisfy*

$$\theta_{R,R';Q} \circ (\tau_{\iota_1}|_Q) \;=\; \tau_{\iota_2}|_Q,$$

where $\theta_{R,R';Q}$ *is the canonical ORA isomorphism of Paper AD; and*

(d) ***Canonicality****:* τ_ι *is the unique* **POT***-morphism* $\mathcal{F}_R \to \mathcal{F}_{R'}$ *over* ι *satisfying (a)–(c).*

Proof. **Existence.** Define τ_ι by assembling the component maps of Lemmas 2.2–2.4. Each component satisfies its respective compatibility condition. The four components together constitute a **POT**-morphism by Definition AA-3.3 (morphisms must preserve all fiber components).

Property (a): The identity enlargement id_R maps Ω_R to itself by the identity injection. Each component map is therefore the identity.

Property (b): For composable enlargements $\iota : R \hookrightarrow R'$ and $\kappa : R' \hookrightarrow R''$, the component map $(\tau_{\kappa\circ\iota})_\Sigma$ sends $[\omega]_R \mapsto [(\kappa \circ \iota)(\omega)]_{R''} = \kappa(\iota([\omega]_R))$, which is the composite of $(\tau_\iota)_\Sigma$ and $(\tau_\kappa)_\Sigma$. The same holds for each other component. Defect-ledger additivity is used for the Λ-component: $\delta_{\kappa\circ\iota}(h) = \delta_\kappa(\iota(h)) + \delta_\iota(h)$, which is the correct composition law.

Property (c): This is the content of Lemma AD's overlap-compatibility conditions in Lemmas 2.2–2.4.

Canonicality (property (d)). Suppose $\phi : \mathcal{F}_R \to \mathcal{F}_{R'}$ is another **POT**-morphism over ι satisfying (a)–(c). Then ϕ is in particular a **POT**-morphism, so it preserves all fiber components. The quotient component ϕ_Σ must send $[\omega]_R$ to a class in $\Sigma_{R'}$ that contains $\iota(\omega)$, because ϕ lies over ι (i.e., it is compatible with the base-category morphism ι). But there is only one class in $\Sigma_{R'}$ containing $\iota(\omega)$, namely $[\iota(\omega)]_{R'}$. So $\phi_\Sigma = (\tau_\iota)_\Sigma$.

The same reasoning forces $\phi_\Lambda = (\tau_\iota)_\Lambda$ (the continuation history of ω under ι is uniquely determined by where ι sends each element), $\phi_T = (\tau_\iota)_T$ (the boundary-law data is determined by the quotient data and the LS-2 structure), and $\phi_D = (\tau_\iota)_D$. Therefore $\phi = \tau_\iota$. □

4 The persistence diagram

Definition 4.1 (Persistence diagram)**.** The *persistence diagram* $\mathbf{D}_{\text{pers}}$ is the functor

$$\mathbf{D}_{\text{pers}} : \mathbf{Reg} \longrightarrow \mathbf{POT}$$

sending each regime R to its **POT**-fiber $\mathcal{F}_R$ and each admissible enlargement $\iota : R \hookrightarrow R'$ to the canonical transition map τ_ι.

Proposition 4.2 ($\mathbf{D}_{\text{pers}}$ is a functor)**.** *The persistence diagram $\mathbf{D}_{\text{pers}}$ is a well-defined functor.*

Proof. Functoriality requires $\mathbf{D}_{\text{pers}}(\text{id}_R) = \text{id}_{\mathcal{F}_R}$ and $\mathbf{D}_{\text{pers}}(\kappa \circ \iota) = \mathbf{D}_{\text{pers}}(\kappa) \circ \mathbf{D}_{\text{pers}}(\iota)$. These are exactly properties (a) and (b) of Theorem 3.1. □

Remark 4.3 (What the persistence diagram is). $\mathbf{D}_{\text{pers}}$ is a directed diagram of **POT**-objects and canonical transition maps, indexed by all admissible regimes and their enlargements. It is the mathematical expression of the claim that the suite's fiber data is not just locally coherent (Paper AD) but globally assembled into a system that can be taken to a limit. The persistence diagram is the object whose universal completion — if it exists — is the universal object $\mathcal{U}$.

5 Consequences for the transport invariant

Corollary 5.1 (Transport invariant is a natural transformation)**.** *The compression-ledger transport invariant $I : R \mapsto I(R, U) = \log|O_R(U)| + \lambda\Lambda_R(U) + \mu \log|C_R(U)|$ defines a natural transformation from $\mathbf{D}_{\text{pers}}$ to the constant functor on $\mathbb{R}_{\geq 0}$, in the sense that for every admissible enlargement $\iota : R \hookrightarrow R'$ and every region U in R:*

$$I(R', \iota(U)) \;=\; I(R, U) \;+\; \delta_\iota(U),$$

where $\delta_\iota(U) \geq 0$ is the defect increment of ι on U (zero for right-stable enlargements).

Proof. By Theorem 3.1, the transition map τ_ι sends $O_R(U)$ to $O_{R'}(\iota(U))$ injectively (no outcomes are lost), $\Lambda_R(U)$ to $\Lambda_{R'}(\iota(U)) = \Lambda_R(U) + \delta_\iota(U)$ (defect additivity), and $C_R(U)$ to $C_{R'}(\iota(U))$ (collar data transported canonically). Substituting:

$$I(R', \iota(U)) = \log|O_{R'}(\iota(U))| + \lambda(\Lambda_R(U) + \delta_\iota(U)) + \mu \log|C_{R'}(\iota(U))| = I(R, U) + \lambda\,\delta_\iota(U). \quad \square$$

Remark 5.2 (Fixing λ: the composition-additivity condition). Corollary 5.1 shows that I changes by $\lambda\,\delta_\iota$ under an admissible enlargement. The composition-additivity condition for λ (Paper AA Step 3) is precisely the requirement that I is additive under composed enlargements:

$$I(R'', (\kappa \circ \iota)(U)) \;=\; I(R, U) \;+\; \lambda\,\delta_\kappa(\iota(U)) \;+\; \lambda\,\delta_\iota(U).$$

This holds automatically from Corollary 5.1 applied twice, for any fixed $\lambda \geq 0$. So λ is *not* forced to a unique value by composition- additivity alone; it can be any non-negative constant. Uniqueness of λ requires a normalization condition beyond composition-additivity — this is the content of the next paper (Paper AF).

6 What the CTM theorem does not prove

Proposition 6.1 (Non-consequences of CTM)**.** *The Coherent Transition-Map Theorem does not prove:*

(i) *that the normalization (λ, μ, ν) is uniquely determined (Paper AF, Open);*

(ii) *that the persistence diagram $\mathbf{D}_{\mathrm{pers}}$ has a colimit in* **POT** *(still requires the universal-completion argument of Paper AA Step 6); or*

(iii) *that any sector, matter, or gauge content is persistent, projected, or realized.*

The CTM theorem produces the diagram. It does not produce its limit.

7 Status ledger update

Label	Claim	Status
AA4	Coherent Transition-Map Theorem	**Unc** *(proved here)*
AE1	Canonical quotient transition map for each admissible enlargement	Unc
AE2	Canonical continuation/ledger transition map	Unc
AE3	Canonical boundary-law transition map	Unc
AE4	Persistence diagram $\mathbf{D}_{\mathrm{pers}} : \mathbf{Reg} \to \mathbf{POT}$ is a well-defined functor	Unc
AE5	Transport invariant is a natural transformation on $\mathbf{D}_{\mathrm{pers}}$ (additive under enlargement)	Unc
AA5	Uniqueness of normalization (λ, μ, ν)	Open *(next target)*

8 Conclusion

The Coherent Transition-Map Theorem is proved. For every admissible enlargement $\iota : R \hookrightarrow R'$ there is a unique canonical **POT**-morphism τ_ι that is the identity on identity enlargements, composes correctly for chains, and respects the ORA isomorphisms on all compatible overlaps.

Together with Paper AD's Overlap Route-Agreement Lemma, this result converts the **POT**-fibration from *locally coherent* to *globally directed*: the fiber data over all admissible regimes assembles into a single functor $\mathbf{D}_{\mathrm{pers}} : \mathbf{Reg} \to \mathbf{POT}$ whose image is a directed diagram of canonically connected **POT**-objects.

The persistence diagram $\mathbf{D}_{\mathrm{pers}}$ is now a mathematical object. The question of whether it has a universal completion — the universal object $\mathcal{U} \in \mathbf{POT}$ — is now a legitimate construction problem in category theory, not a philosophical aspiration.

Two obligations remain before that construction can proceed: uniqueness of the transport invariant normalization (Paper AF), and the amplified minimal-carrier selection theorem (Paper AG). These are the next two papers on the theorem ladder. The ladder is no longer aspirational. Each rung has been verified and the next is precisely named.